

Session 4 — Linear Algebra II

Solution sheet

Complete worked solutions, with reminders of the key ideas. Numerical answers are boxed. Keep tying eigenvalues back to their two big uses: PCA and the stability of dynamical systems.

Exercise 1 $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Solution.

(a) $A(1, 1) = (4 + 1, 2 + 3) = (5, 5) = 5(1, 1)$, so $(1, 1)$ is an eigenvector with eigenvalue $\boxed{5}$.

(b) An eigenvector is a direction the matrix only *stretches* (by the eigenvalue), without rotating it off its own line.

Exercise 2 Eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Solution. $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$. Factoring, $(\lambda - 5)(\lambda - 2) = 0$, so $\lambda = \boxed{5 \text{ and } 2}$.

Reminder

For a 2×2 matrix the characteristic equation is always $\lambda^2 - (\text{trace})\lambda + \det = 0$.

Exercise 3 Eigenvector for $\lambda = 2$.

Solution. $A - 2I = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$, giving $2x + y = 0$, i.e. $y = -2x$. An eigenvector is $\boxed{(1, -2)}$.

Exercise 4 Trace/determinant checks.

Solution. $\text{trace} = 4 + 3 = 7 = 5 + 2 = \lambda_1 + \lambda_2$; $\det = 4(3) - 1(2) = 10 = 5 \cdot 2 = \lambda_1 \lambda_2$. Both check out.

Reminder

Always verify eigenvalues with $\sum \lambda_i = \text{trace}$ and $\prod \lambda_i = \det$ — a free way to catch algebra slips.

Exercise 5 $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ (symmetric).

Solution.

(a) $(3 - \lambda)^2 - 1 = 0 \Rightarrow 3 - \lambda = \pm 1 \Rightarrow \lambda = \boxed{4 \text{ and } 2}$.

- (b) $\lambda = 4$: $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \Rightarrow (1, 1)$; $\lambda = 2$: $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow (1, -1)$. Check $(1, 1) \cdot (1, -1) = 0$: orthogonal.
- (c) Symmetric matrices are guaranteed real eigenvalues and orthogonal eigenvectors (the spectral theorem).

Reminder

Every covariance matrix is symmetric, so its principal axes are always orthogonal — exactly what makes PCA well-behaved.

Exercise 6 Discrete system; eigenvalues 0.9 (dir $\mathbf{u}$) and 0.5 (dir $\mathbf{v}$), each coefficient 1.

Solution. Along an eigenvector, one step multiplies the coefficient by λ .

- (a) After 1 step: 0.9 and 0.5.
- (b) After 10 steps: $0.9^{10} \approx 0.349$ and $0.5^{10} \approx 0.001$.
- (c) The $\lambda = 0.9$ mode dominates: it decays far more slowly, so after many steps the state points essentially along $\mathbf{u}$.

Reminder

A^k multiplies each eigen-coefficient by λ^k . The eigenvalue of largest magnitude sets the long-run behaviour.

Exercise 7 $C = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$.

Solution.

- (a) $(5 - \lambda)^2 - 4 = 0 \Rightarrow 5 - \lambda = \pm 2 \Rightarrow \lambda = 7 \text{ and } 3$.
- (b) PC1 (for $\lambda = 7$): $\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \Rightarrow (1, 1)$.
- (c) Fraction = $\frac{7}{7+3} = 70\%$.

Reminder

PCA is the eigen-decomposition of the covariance matrix: eigenvectors are the principal components, eigenvalues are the variances along them.

Exercise 8 Eigenvalues 12, 6, 1.5, 0.5.

Solution.

- (a) Total = $12 + 6 + 1.5 + 0.5 = 20$; PC1 explains $\frac{12}{20} = 60\%$.
- (b) Cumulative: $\frac{12}{20} = 60\%$, then $\frac{12+6}{20} = 90\%$. So 2 components reach $\geq 90\%$.

Exercise 9 *Eigenvalues 8, 5, 2.*

Solution.

- (a) Total variance = $8 + 5 + 2 = 15$; it equals the *trace* of the covariance matrix.
- (b) No: a covariance matrix is positive-semidefinite, so all eigenvalues are ≥ 0 (a variance cannot be negative).

Reminder

Total variance = $\sum \lambda_i = \text{trace}(\mathbf{C})$, and every eigenvalue of a covariance matrix is non-negative.

Exercise 10 *Could these be covariance matrices?*

Solution.

- (a) Eigenvalues 3, 1: **Yes** — both positive (positive-definite).
- (b) Eigenvalues 4, -2: **No** — a negative eigenvalue is impossible.
- (c) Eigenvalues 2, 0: **Yes, but degenerate** — positive-semidefinite, with zero variance along one direction (the data lie on a line).

Reminder

“Valid covariance matrix” $\Leftrightarrow$ symmetric and positive-semidefinite (all $\lambda \geq 0$). A zero eigenvalue signals a redundant, rank-deficient dataset.

Exercise 11 *Recurrent network, eigenvalues 0.95 and 1.05.*

Solution.

- (a) **Unstable**: $|1.05| > 1$.
- (b) Activity grows without bound along the 1.05 mode (the 0.95 mode decays).
- (c) If both were 0.95: **stable** — every $|\lambda| < 1$, so activity decays to zero.

Reminder

Discrete-time stability requires *every* eigenvalue to satisfy $|\lambda| < 1$. A single mode with $|\lambda| > 1$ makes the whole system blow up.

Exercise 12 *Continuous system, eigenvalues $-1 \pm 2i$.*

Solution.

- (a) **Stable**: the solution behaves like $e^{(-1 \pm 2i)t}$, and the *real part* $-1 < 0$ makes the amplitude decay.
- (b) The imaginary part $\pm 2i$ produces *oscillations* — here, damped oscillations that spiral in toward the equilibrium.

Reminder

Continuous-time stability requires every eigenvalue to have negative real part. Imaginary parts set the oscillation frequency; real parts set growth or decay.

Exercise 13 Continuous system, eigenvalues 2 (*dir* $\mathbf{u}$) and -3 (*dir* $\mathbf{v}$).

Solution. Along $\mathbf{u}$ the state grows like e^{2t} ; along $\mathbf{v}$ it decays like e^{-3t} . After a long time the state aligns with $\mathbf{u}$ and grows without bound.

Exercise 14 100 neurons; first 3 PCs explain 85%.

Solution.

- (a) The population activity is effectively *low-dimensional*: roughly 3 directions capture most of it, even though 100 neurons were recorded.
- (b) Those few components can be plotted and analysed, and dropping the rest removes noise (denoising / visualisation).
- (c) Variance is not the same as *relevance*: a direction carrying little variance may still be the one that matters for behaviour.

Reminder

Low-dimensional structure is ubiquitous in neural data, but PCA ranks directions by variance — not by how informative they are about the task.

Exercise 15 Symmetric $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^\top$.

Solution.

- (a) The columns of $\mathbf{P}$ are the (orthonormal) *eigenvectors* of $\mathbf{A}$.
- (b) The diagonal entries of $\mathbf{D}$ are the *eigenvalues*.
- (c) Applying $\mathbf{A}$ rotates space onto the eigenvector axes, stretches each axis by its eigenvalue, then rotates back — independent stretches along perpendicular directions.

Reminder

This spectral decomposition is why a symmetric matrix is “just” a set of scalings along orthogonal axes — the geometric picture behind PCA and the SVD.